

Class overview and logistic regression

Ciaran Evans

Welcome to STA 712!

Agenda:

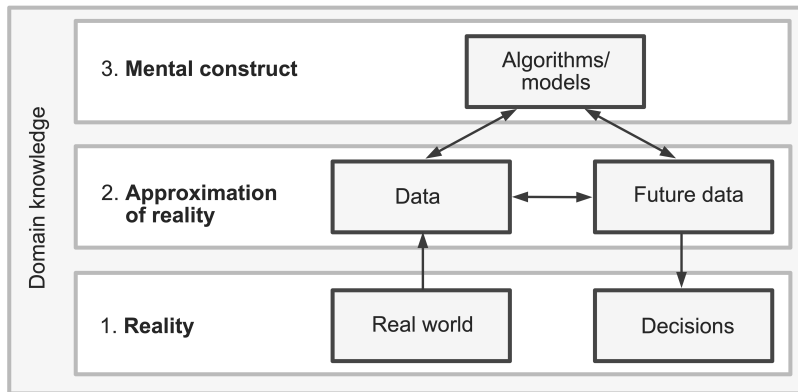
- ▶ Brief course overview and syllabus highlights
- ▶ Sketch plan for first couple weeks
- ▶ Logistic regression

Learning goals

- ▶ **Choosing reasonable methods:** select and fit an appropriate model, and use that model to answer questions of interest
- ▶ **Statistical thinking:** curious about the data, skeptical about the limitations of our methods, and cautious about our conclusions
- ▶ **“What can I do differently?”:** we need to adapt to the data and the situation, not the other way around!
- ▶ **Theory and simulation** to characterize properties and assumptions of statistical models, and explore consequences of different modeling choices
- ▶ **Communicating** results in context

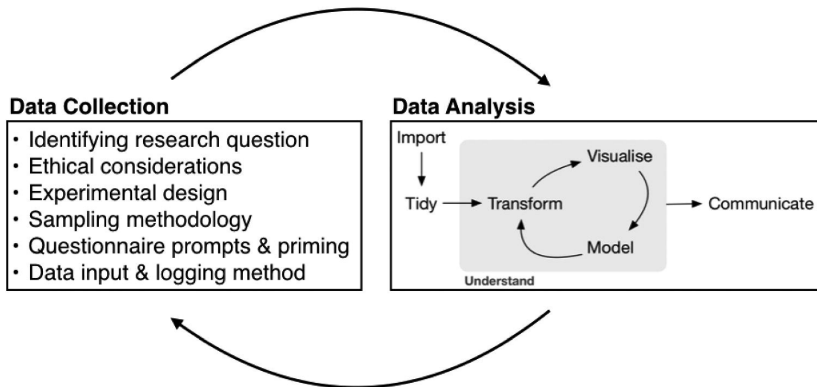
The role of modeling

Essentially, all models are wrong, but some are useful. (Box and Draper, 1987)



(Image from *Veridical Data Science*, Yu and Barter)

Components of (good) data analysis



Albert Y. Kim & Johanna Hardin (2021) "Playing the Whole Game": A Data Collection and Analysis Exercise With Google Calendar, *Journal of Statistics and Data Science Education*, 29:sup1, S51-S60, DOI:

10.1080/10691898.2020.1799728

Overview of course content

- ▶ Core types of models: GLMs (and variants)
 - ▶ Logistic regression (binary responses), Poisson regression (count responses), linear regression (some continuous responses)
 - ▶ GLM theory
- ▶ Model extensions/generalizations
 - ▶ Overdispersion in count data: quasi-Poisson and negative binomial models
 - ▶ Excess 0s: zero-inflated models
 - ▶ Dependent data: mixed models (time permitting)
- ▶ Model assessment
 - ▶ Diagnostics for model assumptions
 - ▶ Assessing model predictions
- ▶ Inference (hypothesis tests and confidence intervals)
 - ▶ Wald and LRT
 - ▶ Parametric bootstrap (time permitting)

Course structure

- ▶ Class participation and seminar attendance
- ▶ Assigned readings
- ▶ Regular HW assignments (due most weeks)
- ▶ Homework presentations (3)
- ▶ One midterm exam
- ▶ Project (throughout second half of semester)
- ▶ Final exam

Plan for the first couple weeks

Recap logistic regression

- ▶ Motivation and interpretation
- ▶ Estimation
- ▶ Inference

Model assessment

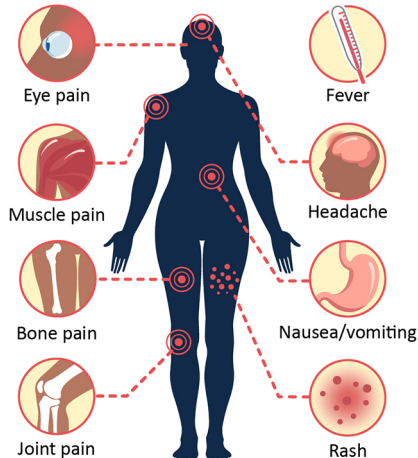
- ▶ Assumptions and diagnostic plots
- ▶ Outliers and multicollinearity
- ▶ Assessing predictions

Motivating example: Dengue fever

Dengue fever: a mosquito-borne viral disease affecting 400 million people a year

Dengue Symptoms

Fever with any of the following



Motivating example: Dengue data

Data: Data on 5720 Vietnamese children, admitted to the hospital with possible dengue fever. Variables include:

- ▶ Sex: patient's sex (female or male)
- ▶ Age: patient's age (in years)
- ▶ WBC: white blood cell count
- ▶ PLT: platelet count
- ▶ other diagnostic variables. . .
- ▶ Dengue: whether the patient has dengue (0 = no, 1 = yes)

Research goal: Predict dengue status using diagnostic measurements

Question: What type of model should we use, and why?

logistic (binary outcome!)
regression

Logistic regression model

$$p_i \in (0, 1)$$

$$\frac{p_i}{1-p_i} \in (0, \infty)$$

$$\log\left(\frac{p_i}{1-p_i}\right) \in (-\infty, \infty)$$

The original authors chose the following model:

$$\underbrace{Dengue_i | Age_i, WBC_i, PLT_i}_{\text{random component}} \sim \text{Bernoulli}(p_i)$$

log-odds

aka
logits

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 \text{Age}_i + \beta_2 \text{WBC}_i + \beta_3 \text{PLT}_i$$

systematic component

Question: Why is there no noise term ε_i in the logistic regression model?

random component captures variability of $Y_i | X_i$

Logistic regression model

"generalized linear model"

```
m1 <- glm(Dengue ~ Age + WBC + PLT,  
           data = dengue, family = binomial)
```

```
summary(m1)$coefficients
```

(response distribution)

##	Estimate	Std. Error	z value	Pr
## (Intercept)	1.252559319	0.1548038204	8.091269	5.90463
## Age	0.138318635	0.0099762568	13.864783	1.03538
## WBC	-0.252329417	0.0135371014	-18.639841	1.52694
## PLT	-0.006027573	0.0006112861	-9.860478	6.17549

Question: How would I interpret the estimated coefficient

$\hat{\beta}_1 = 0.1383$ for Age?

(additive) $0.1383 = \text{change in log odds for 1-year increase in Age (holding WBC \& PLT fixed)}$

(multiplicative) $\exp\{0.1383\} = 1.148 = \text{change in odds for 1-year increase in Age (holding WBC \& PLT fixed)}$
 $= \text{increase of } 14.8\% \text{ in odds}$

"A one-year increase in Age is associated with an increase of 14.8% in odds of dengue, holding WBC \& PLT fixed"

Another example

Examining a health survey in the Netherlands, researchers saw a potential association between having a pet bird and developing lung cancer. They now want to conduct a follow-up study to further examine whether individuals who keep pet birds are at greater risk.

Questions:

- ▶ How would you try to design this study?
- ▶ What data would you want to collect? *Smoking, age, time owning bird, time / amount smoking*
- ▶ What model would you want to use?

*logistic regression
(outcome: lung cancer Y or N)*

*look at coefficient(s) related to bird ownership,
after adjusting for smoking, age, etc.
- interpretation
- test whether $\beta_{\text{bird}} = 0$ vs. $\neq 0$, or > 0 , etc.*

Class activity

```
## (Intercept)          Bird      SexMale      SocEcLow  
## -1.27063830  1.36259456 -0.56127270 -0.10544761 -0.03975  
##      SmokeRate  
##      0.02601689
```

Interpret the estimated coefficient 1.363 in context.

$$\exp\{1.363\} = 3.908$$

- The odds of lung cancer is 3.9 times greater for individuals who keep a bird than for individuals who don't, holding sex, socioeconomic status, age, and smoking habits fixed"

Class activity

```
## (Intercept)          Bird      SexMale      SocEcLow  
## -1.27063830  1.36259456 -0.56127270 -0.10544761 -0.03975  
##      SmokeRate  
##      0.02601689
```

The estimated coefficient on Age is negative. Why might that be the case? Is there anything you would change in the model?

- need to change the function of Age in model (e.g. spline or polynomial)
 - information already captured by smoking history
 - selection effect?
 - could be $\text{Page} \approx 0$
- 

Class activity

```
## (Intercept)          Bird      SexMale      SocEcLow
## -1.27063830   1.36259456 -0.56127270 -0.10544761 -0.03975
##   SmokeRate
##   0.02601689
```

What is the predicted probability of lung cancer for a male individual who keeps a bird, of “high” socioeconomic status, aged 35, who has never smoked?

Class activity

```
## (Intercept)          Bird      SexMale      SocEcLow  
## -1.27063830  1.36259456 -0.56127270 -0.10544761 -0.03975  
##      SmokeRate  
##  0.02601689
```

$$\frac{\exp\{-1.271 + 1.363 - 0.561 - 0.040(35)\}}{1 + \exp\{-1.271 + 1.363 - 0.561 - 0.040(35)\}} = 0.134$$